ESG CLIMATE RISK IN ASEAN FINANCIAL INSTITUTIONS

A Quantitative Assessment of Physical and Transition Risk

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Author: Arif Shaikhan | http://www.linkedin.com/in/arifshaikhan
University: Universiti Kebangsaan Malaysia (UKM)
Course: Actuarial Science
GitHub Repository:
https://github.com/arifshaikhan/ESG-Climate-Risk-ASEAN-Financial-Institutions

1. EXECUTIVE SUMMARY

This research quantifies climate risk exposure for Malaysia's top three banks — Maybank, CIMB, and Public Bank — using NGFS climate scenarios, BNM's CRST 2024 framework, and publicly available loan data.

Key Findings

Metric	Value
Physical risk (flood-exposed loans)	RM168 billion across 3 banks (≈11–12% of portfolios)
Transition risk (carbon price impact)	RM8.5 billion additional ECL under RM150/t CO_2
NGFS stress test range	2.8% to 6.7% portfolio losses on RM100B assets

Strategic Recommendations

1. Mandate geographic disclosure of loan exposures
2. Implement phased carbon pricing (RM50 → RM150/t by 2030)
3. Integrate NETR national policies into NGFS scenarios

Bottom Line

Malaysian banks hold material climate risk that is not currently disclosed. Early action significantly reduces potential losses.

2. PROBLEM STATEMENT

Malaysian financial institutions face increasing regulatory pressure from Bank Negara Malaysia's Climate Risk Stress Testing (CRST 2024) and rising physical risk exposure from floods, temperature increases, and transition risk from carbon pricing.

However, no public analysis has quantified the financial exposure of Malaysian bank loan portfolios under NGFS climate scenarios.

This study addresses this gap by:

- Estimating climate-exposed loan values (physical risk)
- Modelling carbon price impacts (transition risk)
- Stress-testing a representative portfolio under NGFS scenarios

3. METHODOLOGY

3.1 Physical Risk Exposure

Formula:
Physical Risk = Total Loans × High-Risk Sector Share × Flood Exposure %

Assumptions

Sector	Flood Exposure %	Rationale
Property loans	15%	BNM CRST 2024, JPS flood maps
Agriculture	12%	Flood-prone rural states
Manufacturing	8%	Industrial zone exposure

3.2 Transition Risk (Carbon Price)

Formula:
ECL Impact = High-Carbon Loans × Carbon Price × Sensitivity Factor (0.001)

Carbon Price Scenarios

Scenario	Price (RM/t)	Source
Low	50	Voluntary carbon market
Medium	150	BNM CRST 2024 benchmark
High	300	NGFS Net Zero 2050 pathway

High-Carbon Sectors

Manufacturing, Agriculture, Oil & Gas, Utilities, Mining

3.3 NGFS Portfolio Stress Test

A representative RM100 billion portfolio is constructed as follows:

Sector	Weight

Household Mortgages	55%
Manufacturing	12%
Real Estate	10%
Wholesale & Retail	8%
Construction	5%
Agriculture	3%
Oil & Gas / Utilities	3%
Other	4%

NGFS Scenarios

Scenario	Description	Primary Risk
Net Zero 2050	Orderly transition	Transition risk
Divergent Net Zero	Disorderly transition	Transition + physical
NDCs (Hot House)	No additional action	Physical risk

4. DATA SOURCES

Data	Source
Bank loan portfolios	Bursa Malaysia annual reports
Flood zones	JPS Malaysia, BNM CRST 2024
Climate scenarios	NGFS
Carbon prices	BNM CRST 2024, IEA
Regulatory benchmarks	TCFD, BNM Blueprint
Temperature projections	NGFS, MetMalaysia

5. KEY FINDINGS

5.1 Physical Risk — RM168 Billion Exposure

Bank	Exposure	% of Loans
Maybank	RM76.7B	11.5%
Public Bank	RM49.2B	11.6%
CIMB	RM42.5B	9.4%
Total	**RM168.4B**	~11%

Key Insight:
Household mortgages (55–70% of portfolios) are the primary driver. Geographic exposure is not disclosed by banks.

5.2 Transition Risk — RM8.5B+ ECL Impact

Bank	High-Carbon Loans	ECL @ RM150/t
Maybank	RM57.0B	RM8.5B
CIMB	RM57.0B	RM8.6B
Public Bank	RM16.0B	RM2.4B

Key Insight:
Manufacturing is the dominant driver of transition risk. Malaysia currently relies only on shadow carbon pricing.

5.3 NGFS Stress Test Results

Scenario	Loss %	Loss (RM100B)	Primary Risk
Net Zero 2050	2.78%	RM2.78B	Transition
Divergent Net Zero	5.00%	RM5.00B	Transition + Physical
NDCs	6.71%	RM6.71B	Physical

5.4 Malaysia vs Global Peers

Dimension	Malaysia	Singapore	UK/EU	Gap

Carbon pricing	❌ Shadow only	❌ Pending	✅ Yes	Major
TCFD mandate	❌ Coming	❌ Voluntary	✅ Mandatory	Medium
Stress testing	✅ Initial	✅ Active	✅ Mature	Maturity gap

5.5 Bank Climate Readiness

Dimension	Maybank	CIMB	Public Bank
Net zero	Yes (2050)	NZBA member	Not stated
Sustainable finance	RM46.7B	RM117B	Not disclosed
TCFD alignment	Partial	Partial	Limited
Physical risk exposure	RM76.7B	RM42.5B	RM49.2B
Transition risk	RM8.5B	RM8.6B	RM2.4B

6. RECOMMENDATIONS

6.1 Geographic Disclosure

Aspect	Detail
Problem	No disclosure of loan exposure by flood zone
Recommendation	Mandate postcode-level disclosure within 12–18 months
Rationale	Physical risk cannot be managed without spatial data

6.2 Carbon Pricing Framework

Aspect	Detail
Problem	No actual carbon pricing mechanism
Recommendation	RM50 → RM150/t by 2030
Rationale	Shadow pricing does not drive real behaviour change

6.3 Scenario Integration

Aspect	Detail
Problem	NGFS scenarios not tailored to Malaysia
Recommendation	Integrate NETR into stress testing frameworks
Rationale	Improves accuracy of risk modelling

7. IMPLEMENTATION ROADMAP

Horizon	Timeframe	Focus	Key Actions
Horizon 1	0–12 months	Immediate	Disclosure mandate, CRST completion
Horizon 2	1–3 years	Policy	Carbon pricing, TCFD mandate, NETR integration
Horizon 3	3–5+ years	Transformation	Carbon markets, global alignment

8. CONCLUSION

Malaysian banks face material climate risk exposure:

- RM168B in flood-exposed loans
- RM8.5B+ in carbon-related credit losses
- 2.8%–6.7% portfolio losses under NGFS scenarios

Climate risk is financially material but not yet fully priced or disclosed.

Early intervention through disclosure, pricing, and scenario integration can significantly reduce systemic risk and position Malaysia as a regional leader in climate-resilient finance.

9. APPENDIX: ACRONYMS

Acronym	Meaning
BNM	Bank Negara Malaysia
CRST	Climate Risk Stress Testing
ECL	Expected Credit Loss
NGFS	Network for Greening the Financial System
TCFD	Task Force on Climate-related Financial Disclosures
NETR	National Energy Transition Roadmap

10. REFERENCES

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